AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 36 · STUDENT LESSON

Creator Defense

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

State the claim. Show the record. Explain the limit.

Defend a grounded assistant through separate outcomes and an inspectable evidence bundle.

Keep one habit: Claim guide → D1 rates → D2 supported/missing demo → challenge rehearsal → independent outcome defence.

MAIN LESSON · SECTION 1

Can every claim in your defence point to a record?

> Explain the architecture, results and limitations without filling gaps with a confident story.

Your learning goals

I can explain the **mechanism** and terminology of generative systems **defense**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or **limitation** and design a revealing follow-up.

Recall bounded comparison evidence

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Recall Capstone Evaluate and Improve: archive the original comparison, keep mandatory failures visible and distinguish exposed revision cases from new reserved **evidence**. A persuasive defence must not expand those results into unmeasured claims.

Your end goal

By the end, I can explain creator **defense** using a traced example and a stated **limitation**.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Capstone Evaluate and Improve. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Capstone Design. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Bias and Representation. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 36: Creator **Defense**

A creator explains **architecture**, **evidence**, **limitation**s, provenance, and responsible use.
</details>



CREATOR MISSION

Demo and defend against fresh test and design questions.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Organise the defence around evidence-backed claims

> A polished demo is one observation, not a complete evaluation.

Understand a defensible presentation

A creator defence explains the problem, design choices, source boundaries, implementation and evidence to someone who has not seen the project. Demonstrate a supported path and a missing-answer path. Separate retrieval, supported answers, fallback and allowed actions; they are different outcomes. State which results were executed, traced on paper or supplied for practice.

A clear account of a failure and its limits is stronger evidence of understanding than an invented perfect result.

A generative-system **defense** presents the question, **architecture**, corpus, permissions, baseline, evaluation and unresolved failures. Demonstration fluency is only one observable property.

Label fictional practice separately from project results

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use creator **defense** to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

MAIN LESSON · SECTION 3

Attach each headline to the right evidence

> The unit and denominator belong beside the rate.

Complete claim-to-record guide

Defence claim	Matching evidence and limitation
User value	Defined task and baseline comparison; no invented benefit
Architecture	Source/corpus versions, stages, permissions and owner map
Retrieval	Defined relevant-retrieval outcome and its own denominator
Answer support	Claim-level rubric and supported-answer denominator
Fallback	Missing-source cases and appropriate behavior rate

Defence claim	Matching evidence and limitation
Actions and privacy	Actual recorded scope/trace; a zero in one run is not universal protection
Speed/resources	Timing/counts/tariff only when measured and defined
Reproduction	Exact versions/cases/outputs where available; mark missing packet fields
Fresh challenge	Predict, explain and record support; do not rewrite earlier evidence

Link each claim to a recorded result. State the distinction between retrieval relevance, claim support, fallback and permissions. Give a **reproduction packet** and answer challenges without inventing missing measurements.

Respect missing measurements

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the deciding check

- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Demo and defend against fresh test and design questions.
↓
SYSTEM
A creator explains architecture, evidence, limitations, provenance, and responsible use.

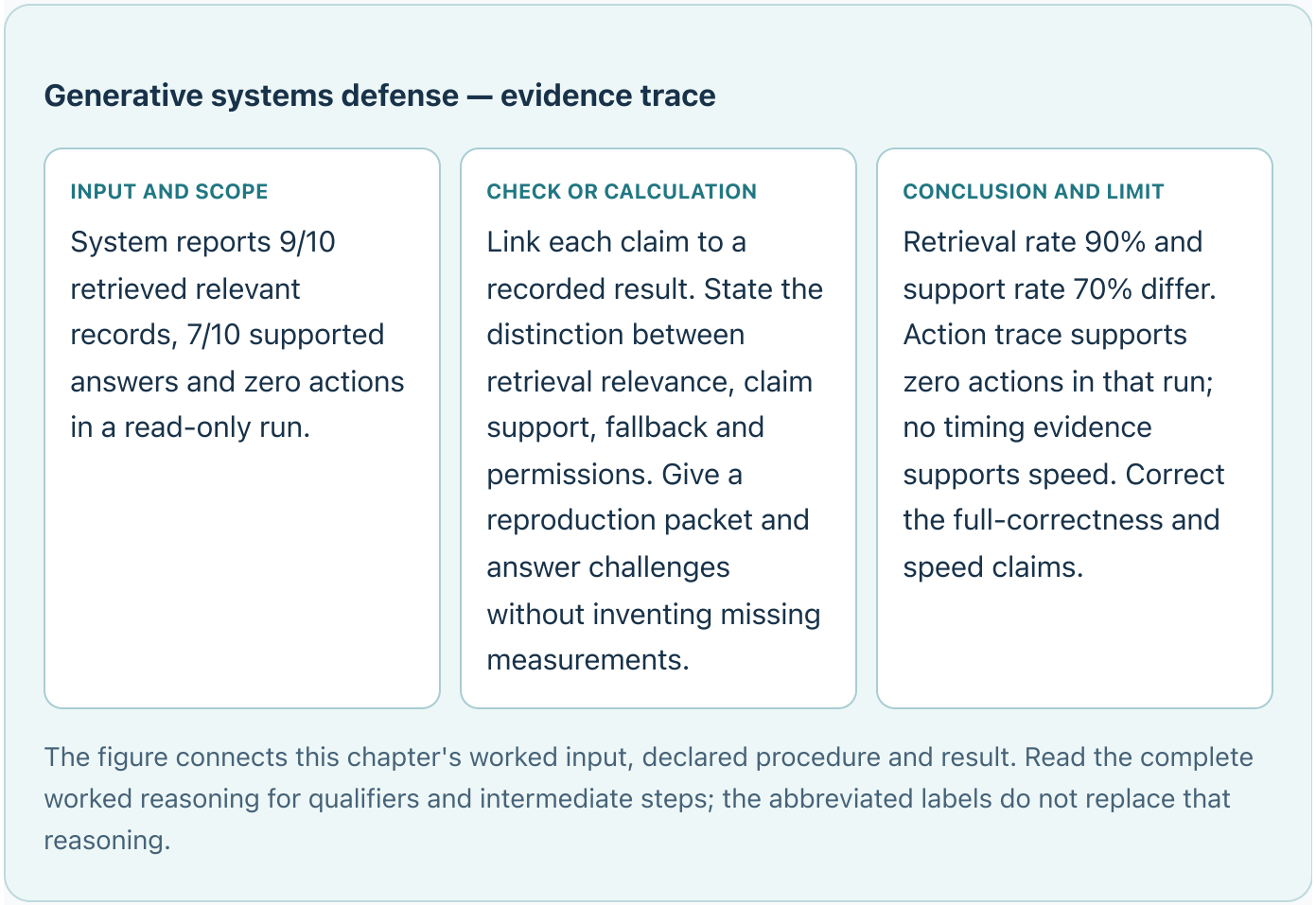
↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

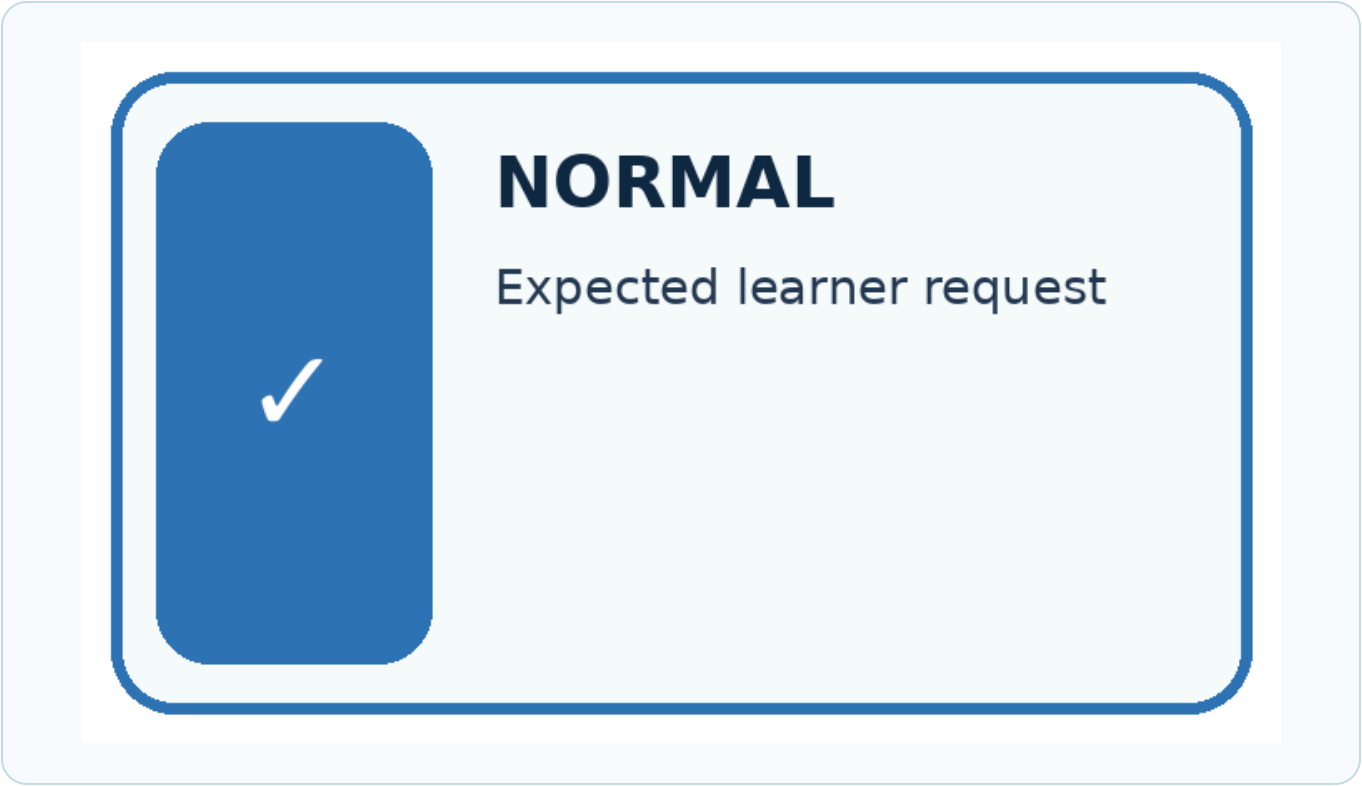
Connect the picture and the explanation

MAIN LESSON · SECTION 4

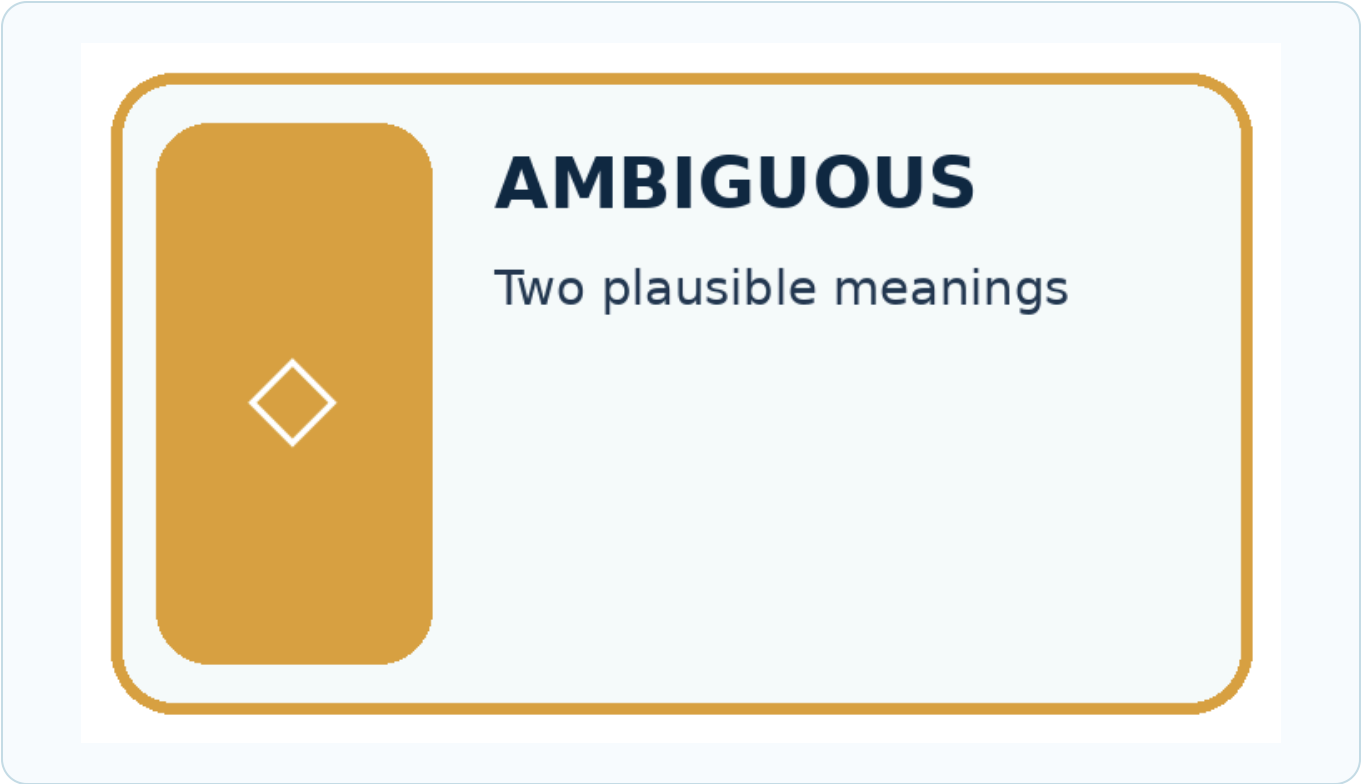
Compare a truthful limitation with an unsupported headline

> Retrieving relevant evidence does not establish correct use in the final answer.

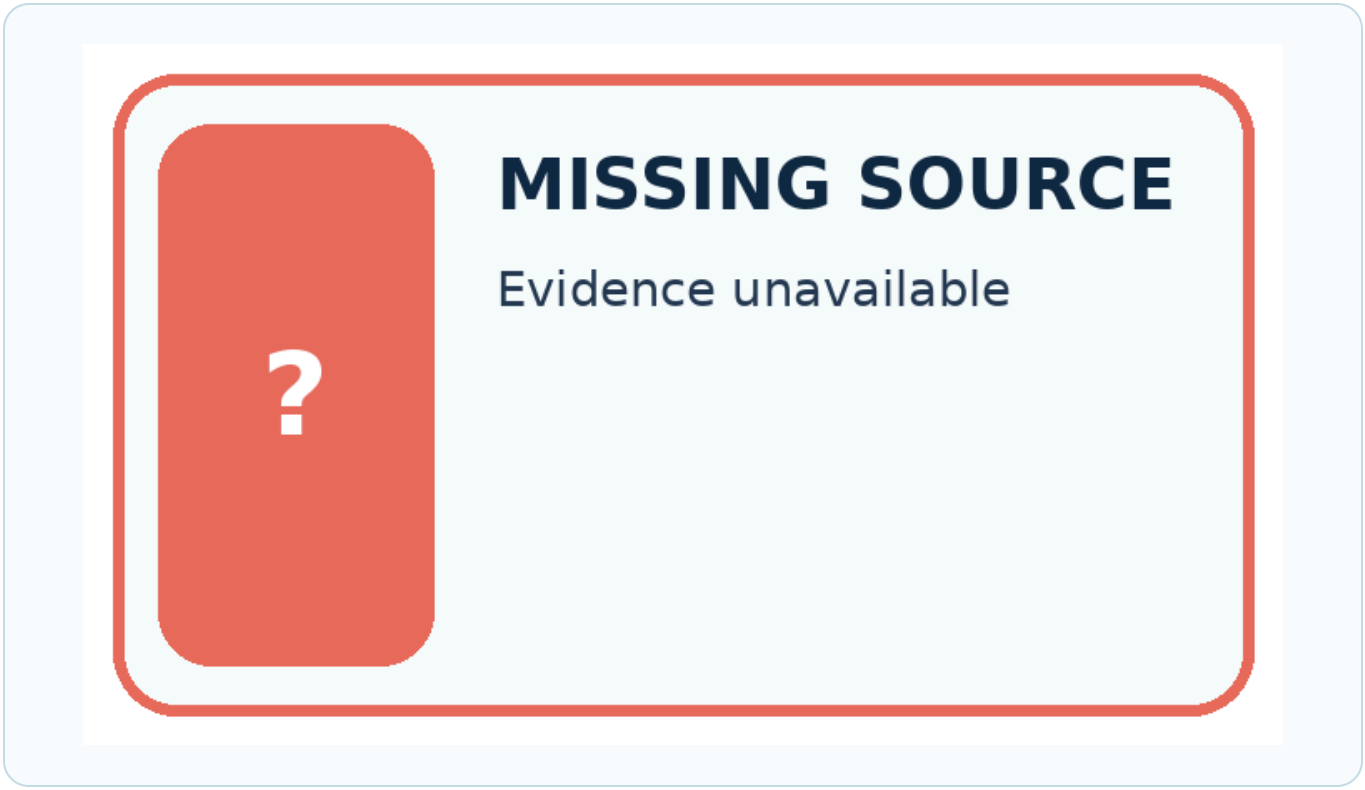
Four pictures with a purpose



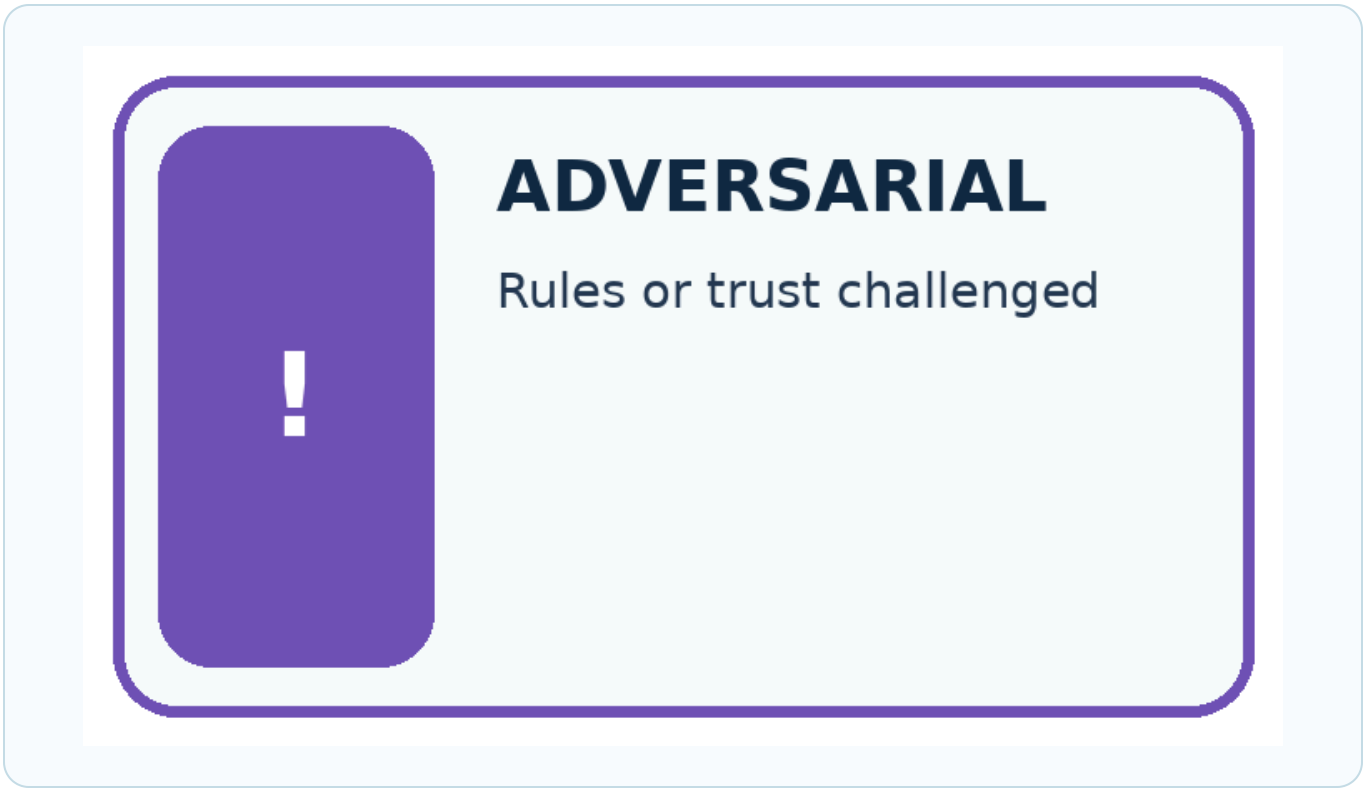
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Only the retrieval rate is put on the presentation poster.

<details>
<summary>Check after predicting</summary>

Prediction: The poster omits the lower supported-answer rate and cannot imply **80%** fully correct answers.

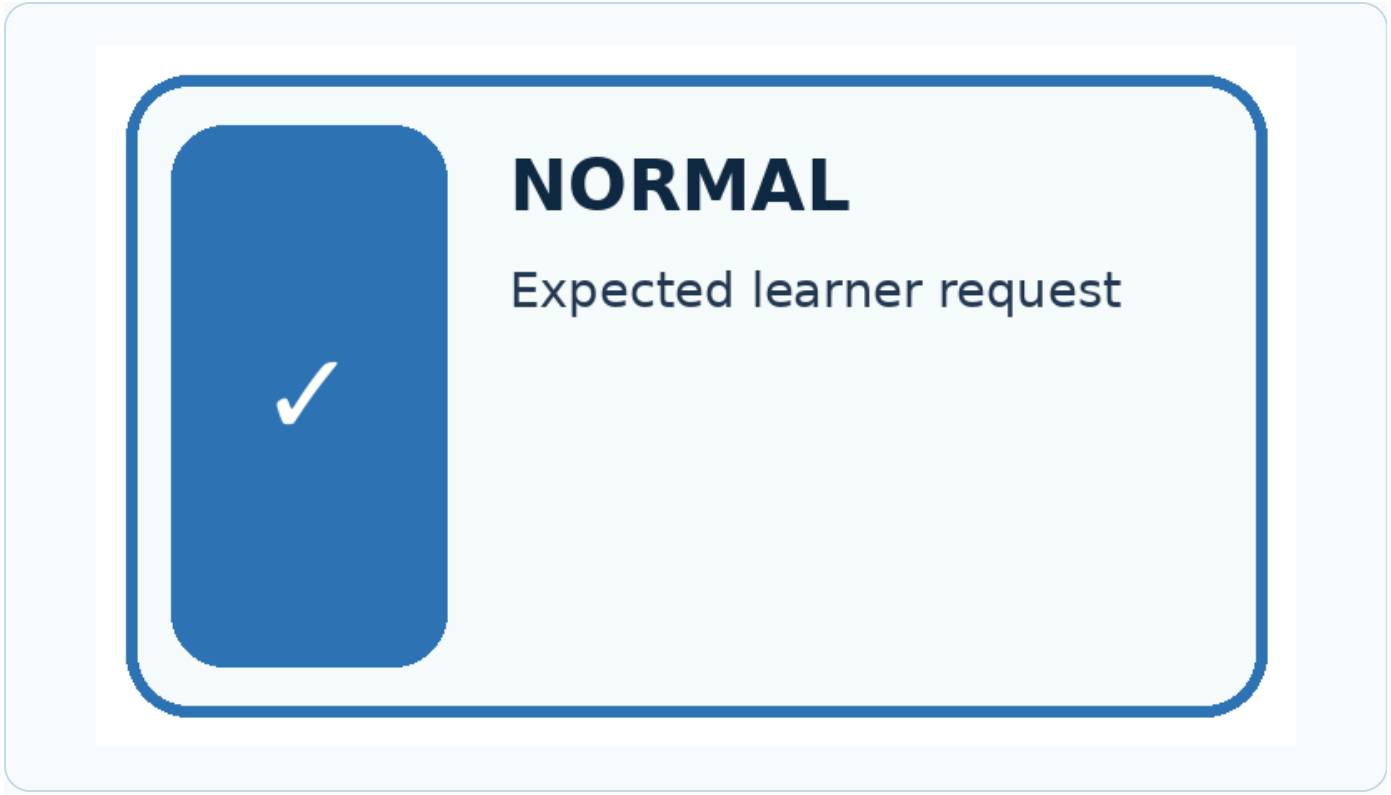
Reason: Retrieving useful **evidence** is an intermediate step, not proof that the final answer uses it correctly.
</details>

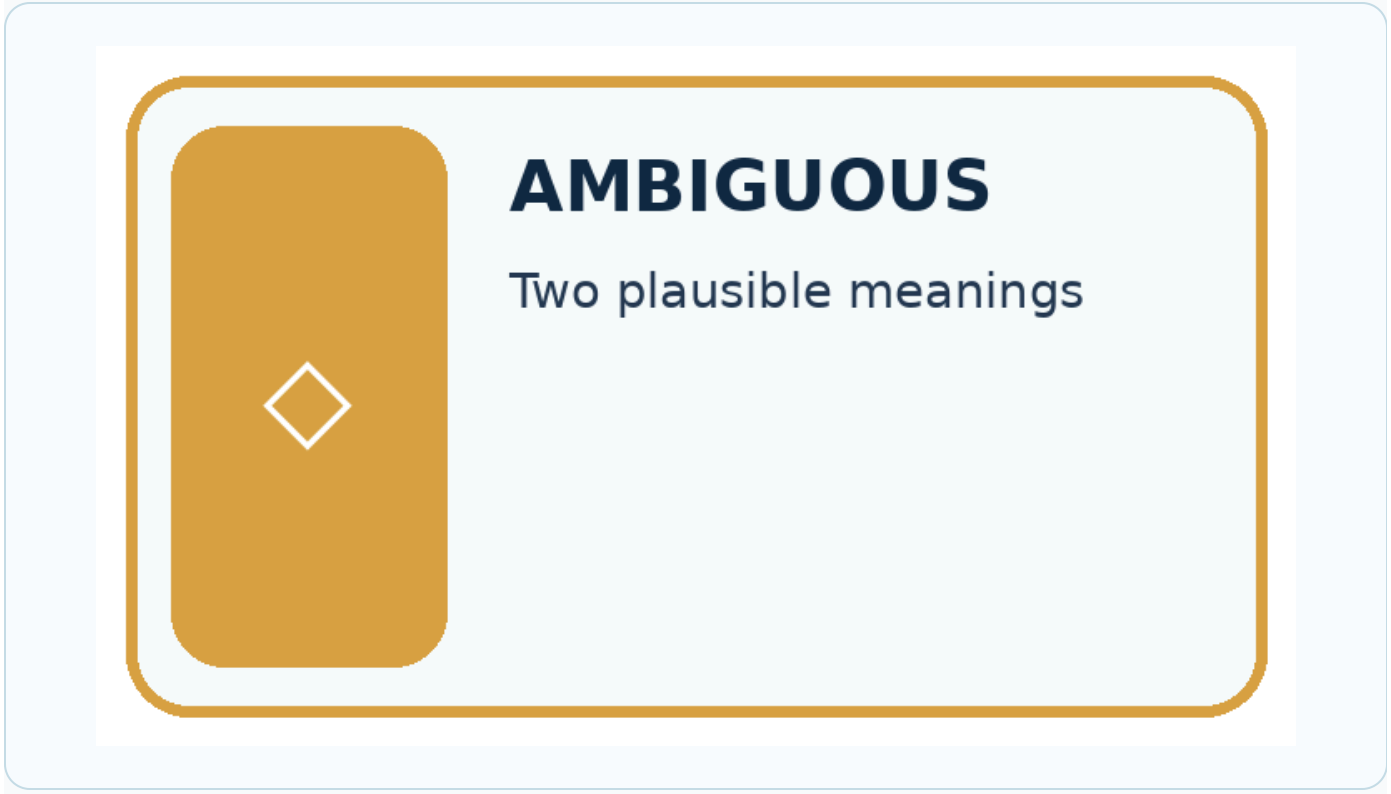
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<summary>Earlier wording, alternatives and picture transcripts</summary>

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.
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NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace three separate claims from practice packet D1

> The honest poster keeps the lower support rate and missing measurements visible.

Read the complete aggregate record and scope

Defend a study-buddy project using separate claims and matching **evidence**.

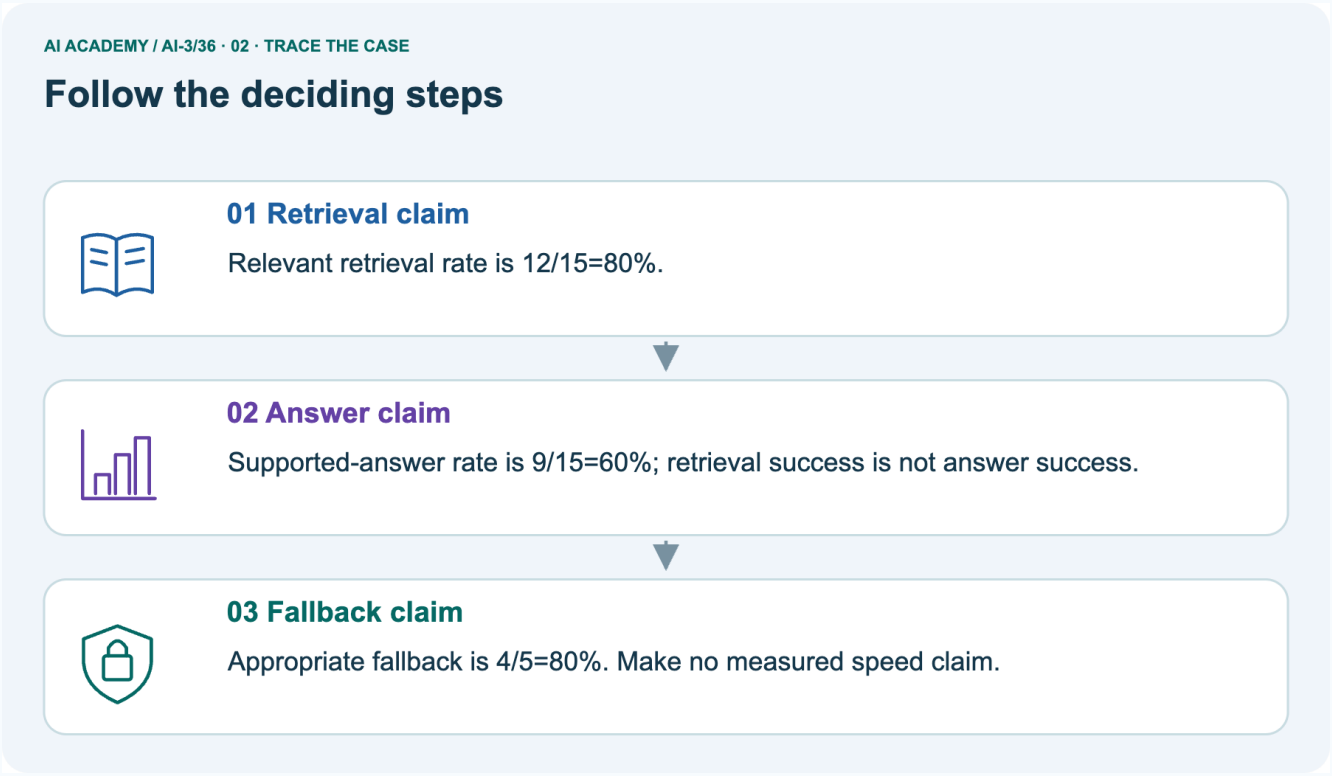
Set A: 15 answerable questions, with 12 relevant retrievals and 9 supported answers. Separate set B: 5 missing-source questions, with 4 appropriate fallbacks. No speed measurements exist.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Practice packet D1 supplies aggregate counts only: 15 answerable questions, 12 with relevant retrieval and 9 with supported answers; separate 5 missing-source questions with 4 appropriate fallbacks. Treat retrieval as a per-question pass indicator in this fixture, not document precision@k. Rates 80%, 60%, 80% answer different questions.

No timing, energy, case-level failure outputs or complete reproduction packet is supplied. Do not invent a specific failure explanation or a speed comparison from these totals.

Trace retrieval, support and fallback denominators



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Retrieval claim	Relevant retrieval rate is 12/15=80%.
2. Answer claim	Supported-answer rate is 9/15=60%; retrieval success is not answer success.
3. Fallback claim	Appropriate fallback is 4/5=80%. Make no measured speed claim.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

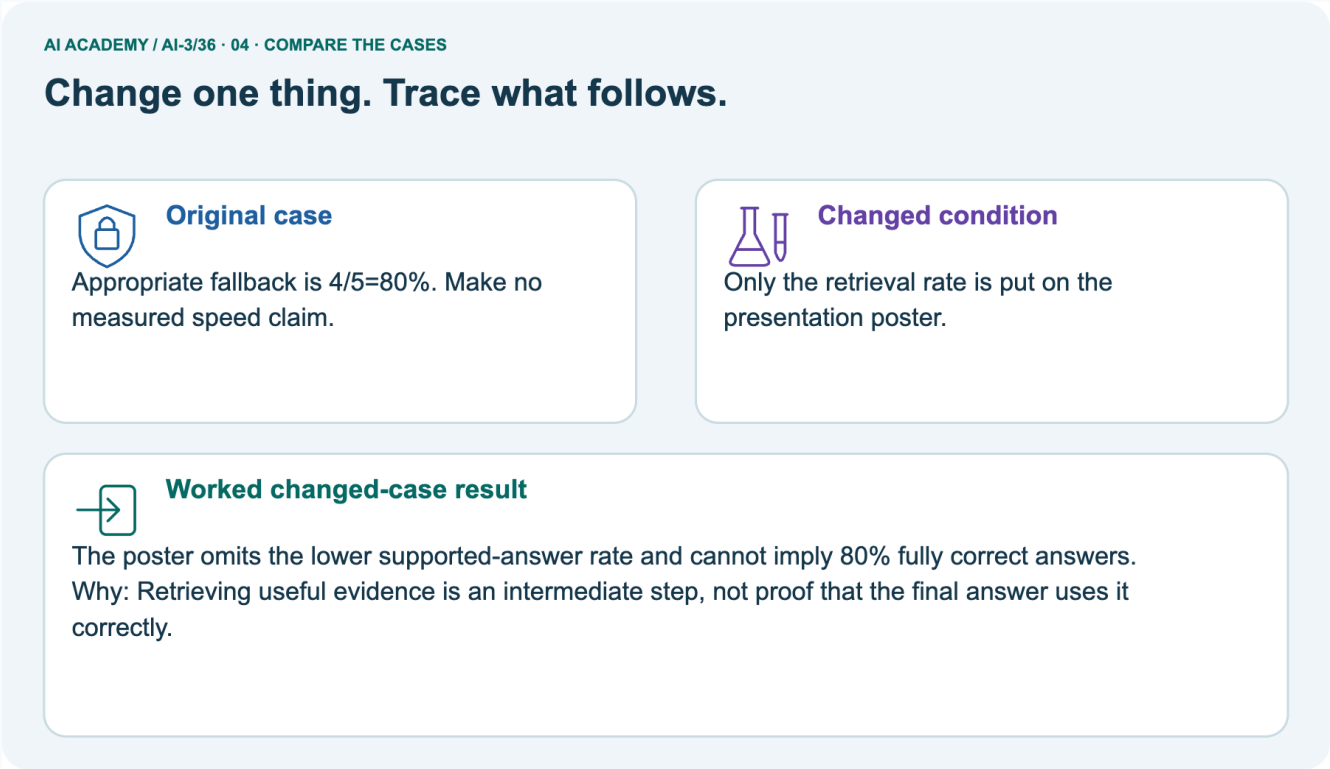
Read the picture in order. Appropriate fallback is 4/5=**80%**. Make no measured speed claim. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict showing only the retrieval rate

- Change one thing: Only the retrieval rate is put on the presentation poster. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

State fictional-count and project-evidence limits

- These invented counts illustrate **defense** structure; the class must replace them with its own documented **evidence**.

- Read the labelled picture: Which numbers would you display to make the **defense** honest? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

▼ Separate practice · Ten-record/answer W2

System reports 9/10 retrieved relevant records, 7/10 supported answers and zero actions in a read-only run. Presenter claims 90% fully correct and faster than baseline.

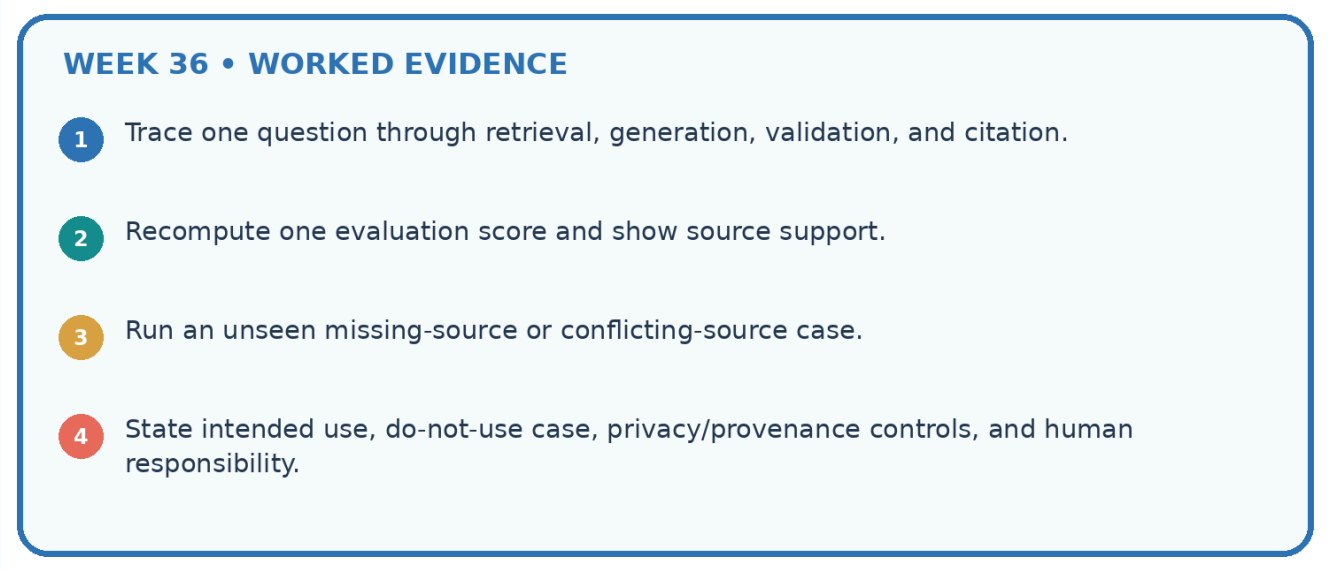
Retrieval rate 90% and support rate **70%** differ. Action trace supports zero actions in that run; no timing **evidence** supports speed. Correct the full-correctness and speed claims.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Trace one question through retrieval, generation, validation, and citation.

2. Recompute one evaluation score and show source support.

3. Run an unseen missing-source or conflicting-source case.

4. State intended use, do-not-use case, privacy/provenance controls, and human responsibility.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

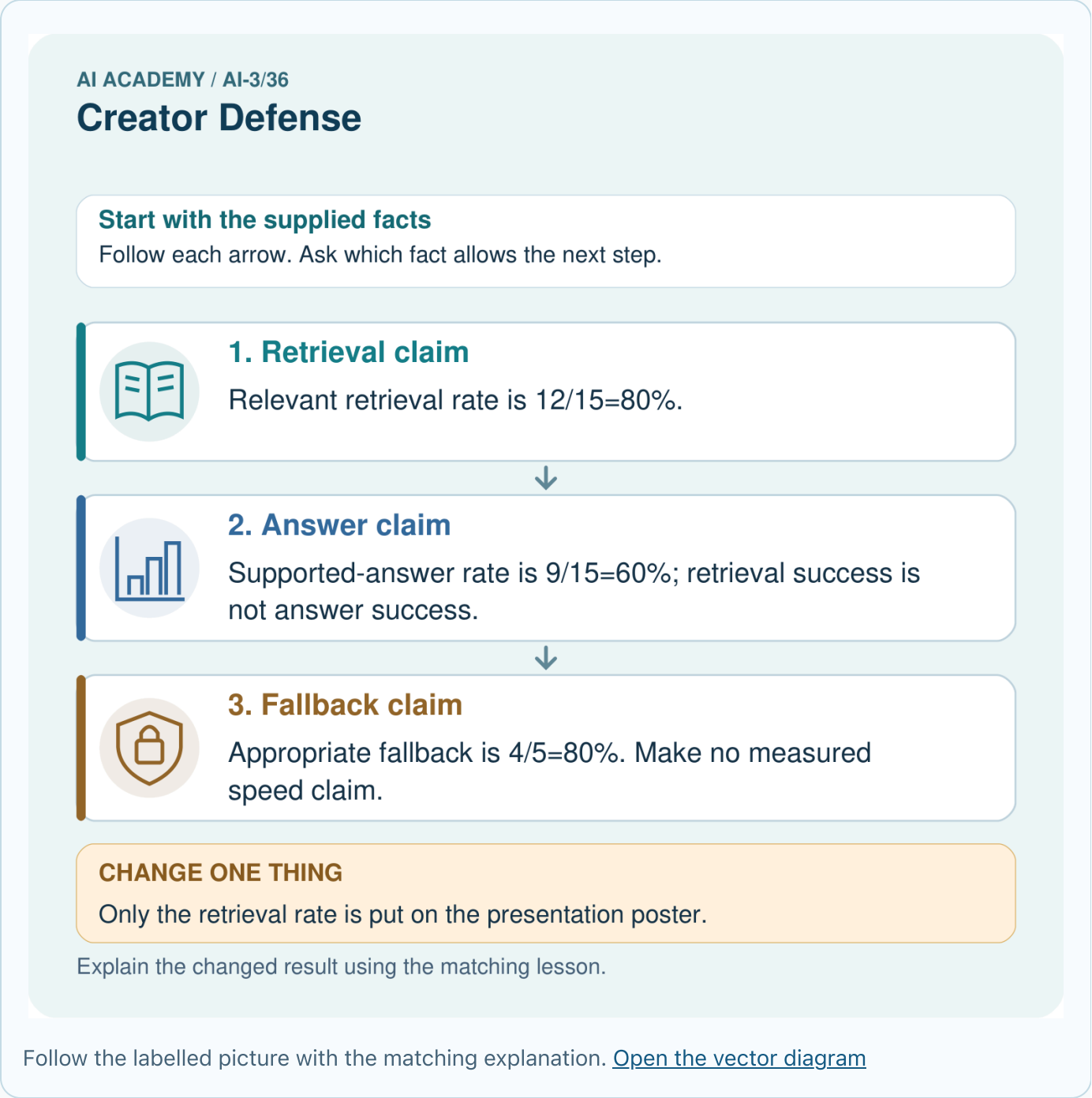
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Creator Defense

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find the missing denominator or unsupported comparison

> Correct the claim instead of changing which evidence is shown.

Repair full-correctness and speed headlines

Claim to repair: The project is 80% fully correct because relevant retrieval was 12/15. Supported answers are 9/15=60% in the supplied answerable set, and fallback is a separate 4/5. Also reject faster than baseline: no timing record exists. Six answerable cases failed the support criterion, but their exact errors are unavailable; request the case-level records instead of inventing a failure narrative.

A missing measurement is a reported limit, not a number to estimate casually.

Design a revealing follow-up

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A smooth demo proves the system is reliable. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

MAIN LESSON · SECTION 8

Practise the separate W2 defence with less help

> Distinguish retrieved-record relevance, answer support and action trace scope.

The demonstration-versus-generalisation limit

A **defense** should preserve negative **evidence** and uncertainty rather than treating polished presentation as proof.

Complete the 9/10 relevance and 7/10 support W2

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Generative systems **defense**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Audit an all-future-cases claim

- Use the worked case for Generative systems **defense**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check your reasoning

Explain the pictured evidence choices

- Which numbers would you display to make the **defense** honest? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Explain the picture

> **My explanation using a pictured step and the given facts**

Keep independent figures separate

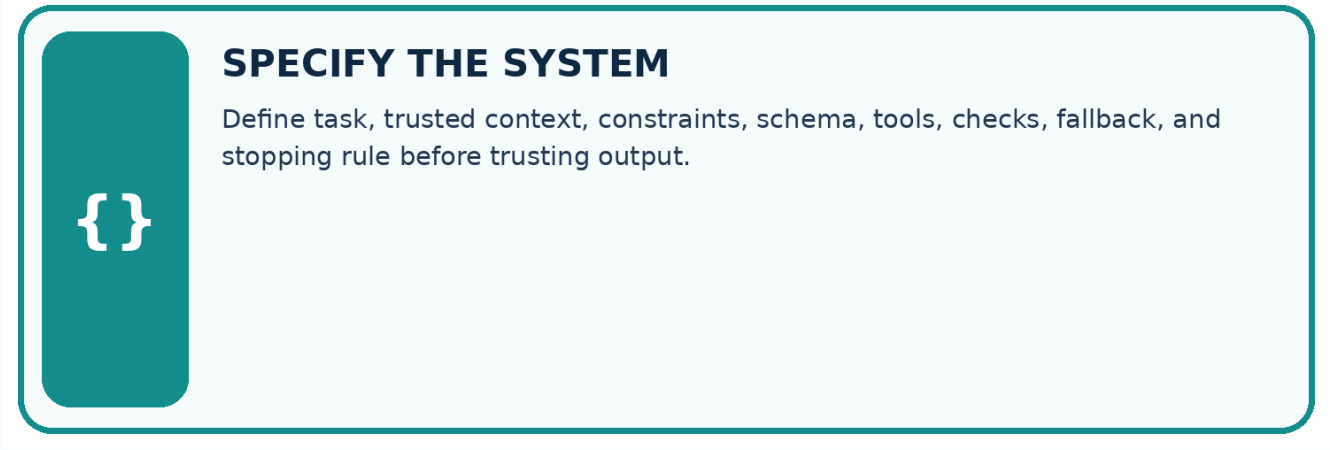
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + **limitation**s

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

MAIN LESSON · SECTION 9

Rehearse with a complete packet and visible gaps

A defence can explain what is known and what the record still cannot establish.

All supplied architecture, source and result records

Complete supplied rehearsal evidence	What may be claimed
Architecture proposal	Question→eligible source→answerability→draft→support check→display; read-only. This is a proposed diagram, not execution evidence.
D1 aggregate counts	15 answerable: 12 relevant retrievals, 9 supported answers; 5 missing: 4 appropriate fallbacks. No case-level outputs or timings.
D2 fresh demo source	Approved fictional club note C1-v1: Club Cedar meets Tuesday. No leader name is supplied.
D2 questions	Which day does Cedar meet? / Who leads Cedar?
D2 expected behavior	Tuesday [C1-v1] / leader unavailable from this note. No actual model output supplied.

Complete supplied rehearsal evidence	What may be claimed
Packet gaps	No exact evaluated build/corpus IDs for D1, individual failures, original outputs or timing. Request them; do not invent reproducibility.

Six evidence-based challenge prompts

Rehearsal challenge	Expected evidence-based response
Why not call D1 fully correct at 80%?	Support is 60%; retrieval and support are different outcomes.
Which path can you demo from D2?	Source-backed day and an honest missing-leader answer; label as paper prediction until executed.
What failed in the six unsupported answer cases?	Exact types unavailable; request case-level records.
Was the system faster or cheaper?	No timing/tariff/count comparison supplied.
Can this packet reproduce the original run?	No; build/source/case/output details are incomplete.
What would you change next?	Choose a diagnosed failure only after inspecting its record; predict a repair and preserve fresh-test roles.

Present, predict a fresh case and preserve limitations

1. Build a short architecture-and-evidence presentation using the complete rehearsal packet. State task, baseline requirement, stages, source limits and the three rates with denominators. Let a partner choose a visible rehearsal challenge, then answer from the available evidence and mark missing facts.

   Treat D2 as a fresh paper scenario relative to D1, not as measured system performance or a hidden assessment once it has been discussed. For your actual project, substitute its documented evidence or mark unavailable; never reuse D1 counts as personal results. Explain one tradeoff as a proposal with its unmeasured consequence. Record what a reviewer would need to reproduce a run.

   Keep the independent 8/10, 6/10, 3/4 case withheld until the first attempt.

Keep the defence evidence

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____

> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

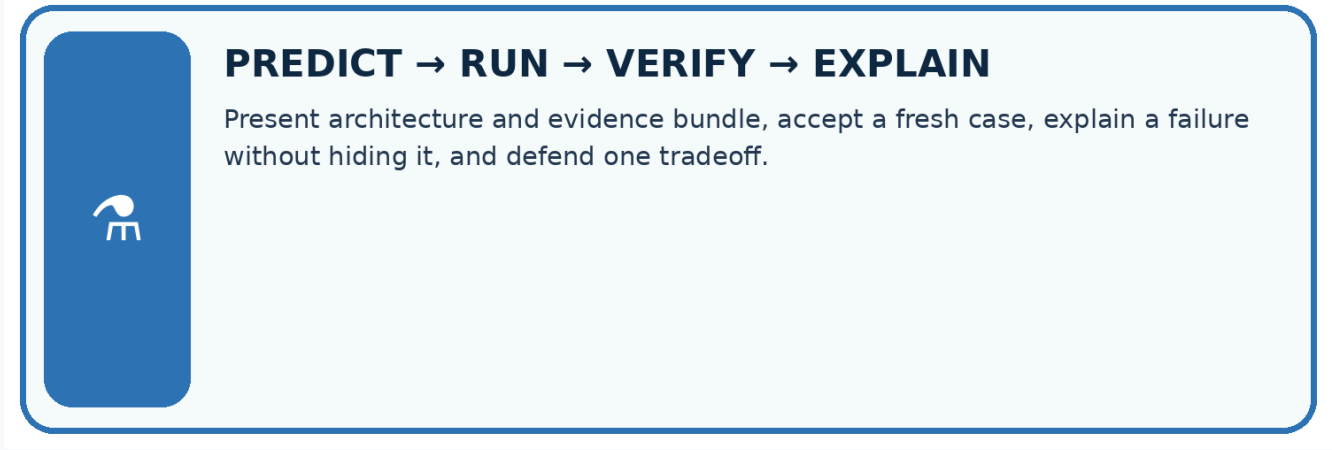
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the complete D1/D2 rehearsal packet and challenge cards. Present separate rates, trace a supplied supported/missing-source scenario, identify missing reproduction details and reject unmeasured speed claims. Label predictions, paper traces and actual project results separately; do not adopt fictional counts as student **evidence**.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: defend a fresh set of outcome figures

> State three rates and identify an unsupported headline.

Independent record and scope

A **defense** has retrieval 8/10, support 6/10, fallback 3/4 and no measured latency. State the three rates and reject one unsupported claim.

Use only the independent rates' stated counts and missing-latency fact. The outcome units and denominators must be named separately. No measured speed, energy comparison or universal correctness result is supplied.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Week 36 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

An audience asks whether your assistant is always correct. Give an **evidence**-based **defense** using results and a limit.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Final product card documents purpose, system map, sources, evals, limits, risks, and human oversight.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Your assistant passes 9/12 final questions, including all six direct facts but only 3/6 ambiguous or absent-**evidence** cases. Report results and argue for a focused next step.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project defence inspectable after the presentation

> Evidence and limits should travel with the shared artefact.

This week’s contribution

For the Verified Study Buddy, finish a versioned evidence bundle with the user need, baseline, architecture, source/permission ledger, implementation, rubric, case roles, outputs, failures and unresolved claims. Keep a simple supported/missing-source demonstration alongside evaluation results. Explain who owns consequential decisions and what the system cannot establish.

Archive the original and revised project records so a reviewer can inspect the tradeoffs rather than only a final poster.

For your Verified Study Buddy, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Retain checkpoint and human choice

THIS WEEK’S CHECKPOINT Final product card documents purpose, system map, sources, evals, limits, risks, and human oversight.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

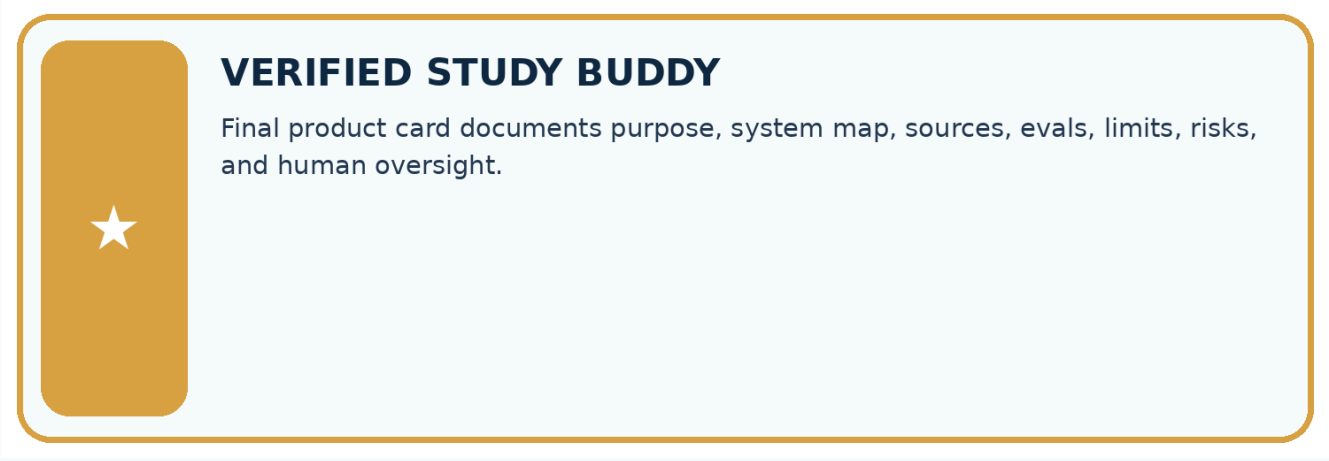
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse fluent **output** with verified truth.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

Finish: state the claim, evidence and honest limit

> A creator can explain a failure without hiding or inventing results.

Explain without notes

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of generative systems **defense**

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
defense	An explanation of a project decision supported by evidence and an account of its limitations.	
architecture	The arrangement of a system's parts and the connections between them.	
limitation	A condition or missing capability that restricts what a result can establish or a system can safely do.	
transfer test	A fresh task that checks whether learning can be applied without the original supports.	

Terms for the connected explanation

Word	Meaning
Defense claim	A statement tied to inspectable project evidence

Word	Meaning
Reproduction packet	Artifacts needed to repeat a reported procedure
Unmeasured outcome	A property for which the experiment supplies no observation

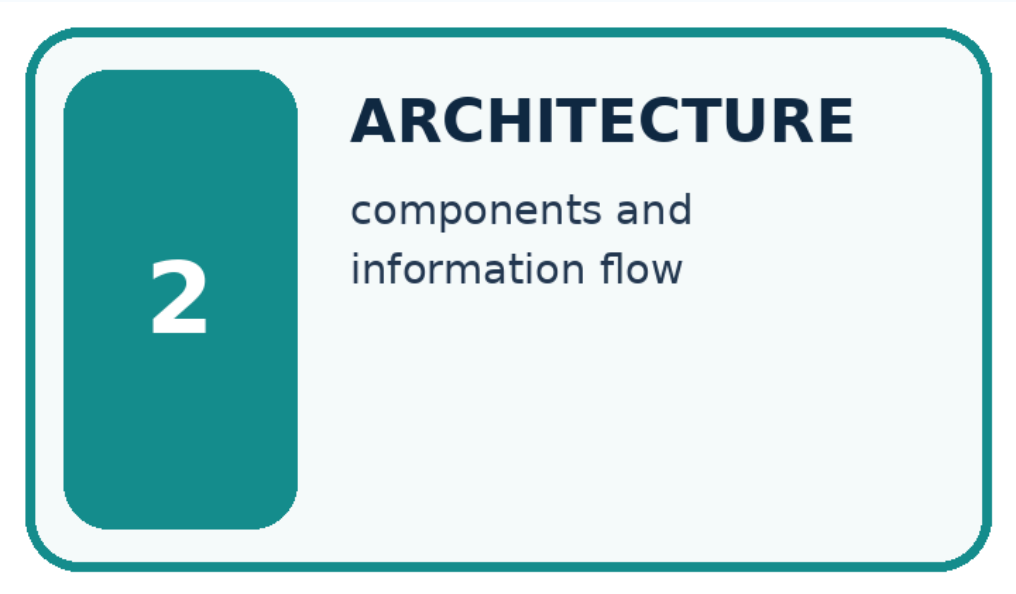
▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
ARCHITECTURE
components and
information flow


3
LIMITATION
condition where
performance weakens



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

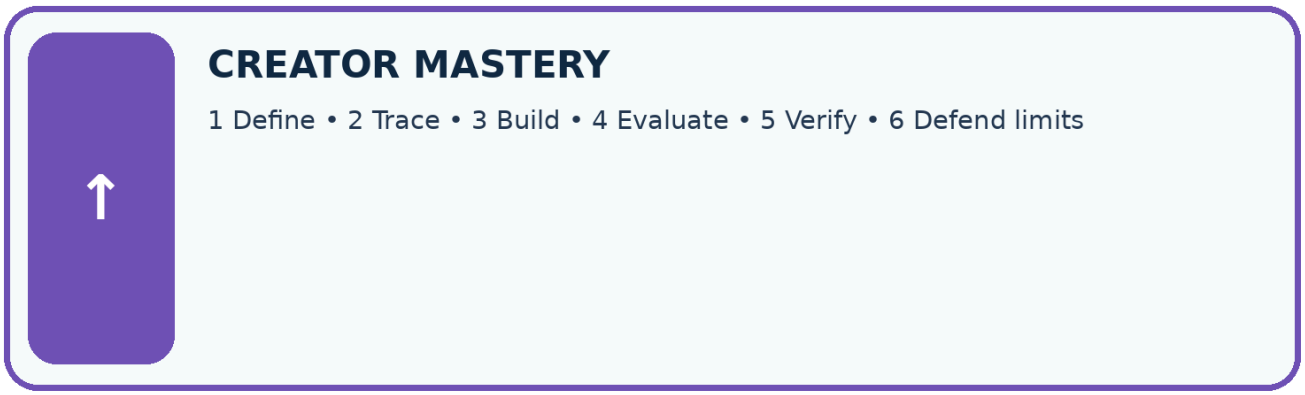
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Defend the fresh outcome figures independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One **limitation**, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

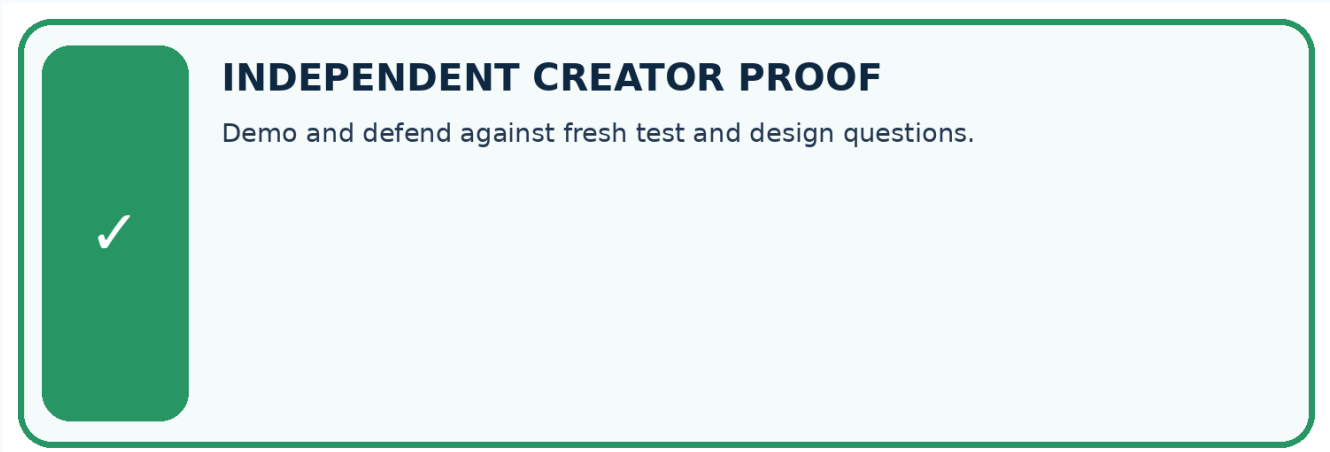
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a **limitation** ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Defend the fresh outcome figures independently is the selected independent case.



EXPLANATION CHECK Require separate rates and denominators, matching evidence and rejection of unmeasured claims.

Write or draw your reasoning and evidence.

▼ **Extra practice: Defend a bounded claim**

REFERENCE · SECTION 13

Extra practice: Defend a bounded claim

> Try the question before opening its worked explanation.

Read the supplied case and question

Ten successful demos lead to a never hallucinates claim.

Write a defensible conclusion.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Report those ten specified passes, show **evidence** and fallback, and state untested conditions. Demonstrate an unanswerable question.

Evidence or stage	Value or result
Result	Ten specified demonstrations passed
Evidence	Show sources and checks
Limit	Other conditions remain untested
Demonstration	Show an unanswerable question

▼ Earlier wording, alternatives and picture transcripts

Week 36 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Defend a bounded claim

Ten successful demos lead to a never hallucinates claim.

Write a defensible conclusion.

Report those ten specified passes, show **evidence** and fallback, and state untested conditions. Demonstrate an unanswerable question.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Explain the mechanism and terminology of generative systems defense	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Defend a study-buddy project using separate claims and matching **evidence**.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- Defend a study-buddy project using separate claims and matching **evidence**.
- Which labelled step matches this detail? Appropriate fallback is 4/5=**80%**. Make no measured speed claim.
- 1. Fallback claim
- 2. Answer claim
- 3. Retrieval claim
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.
- After your attempt, compare with the supported explanation: Fallback claim Appropriate fallback is 4/5=**80%**. Make no measured speed claim. Read the picture in order. Appropriate fallback is 4/5=**80%**.

  Make no measured speed claim. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check the first explanation

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: Only the retrieval rate is put on the presentation poster. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. The poster omits the lower supported-answer rate and cannot imply **80%** fully correct answers.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.
- After your attempt, compare with the supported explanation: The poster omits the lower supported-answer rate and cannot imply **80%** fully correct answers.

  Retrieving useful **evidence** is an intermediate step, not proof that the final answer uses it correctly.

My selected conclusion, deciding fact and explanation

▼ Check the changed case

Repair a missing building block

Repair and save your work

Review Capstone Evaluate and Improve (ai-3/35). Goal: Explain the **mechanism** and terminology of capstone evaluation and revision. Short practice: May the team call the repaired validation cases an untouched final test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Capstone Design (ai-3/33). Goal: Explain the **mechanism** and terminology of capstone requirements and corpus. Short practice: What should the project plan record before the new flower test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Bias and Representation (ai-3/30). Goal: Explain the **mechanism** and terminology of representation and bias evaluation. Short practice: Explain why comparing six correct answers in each group would mislead.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep your completed work

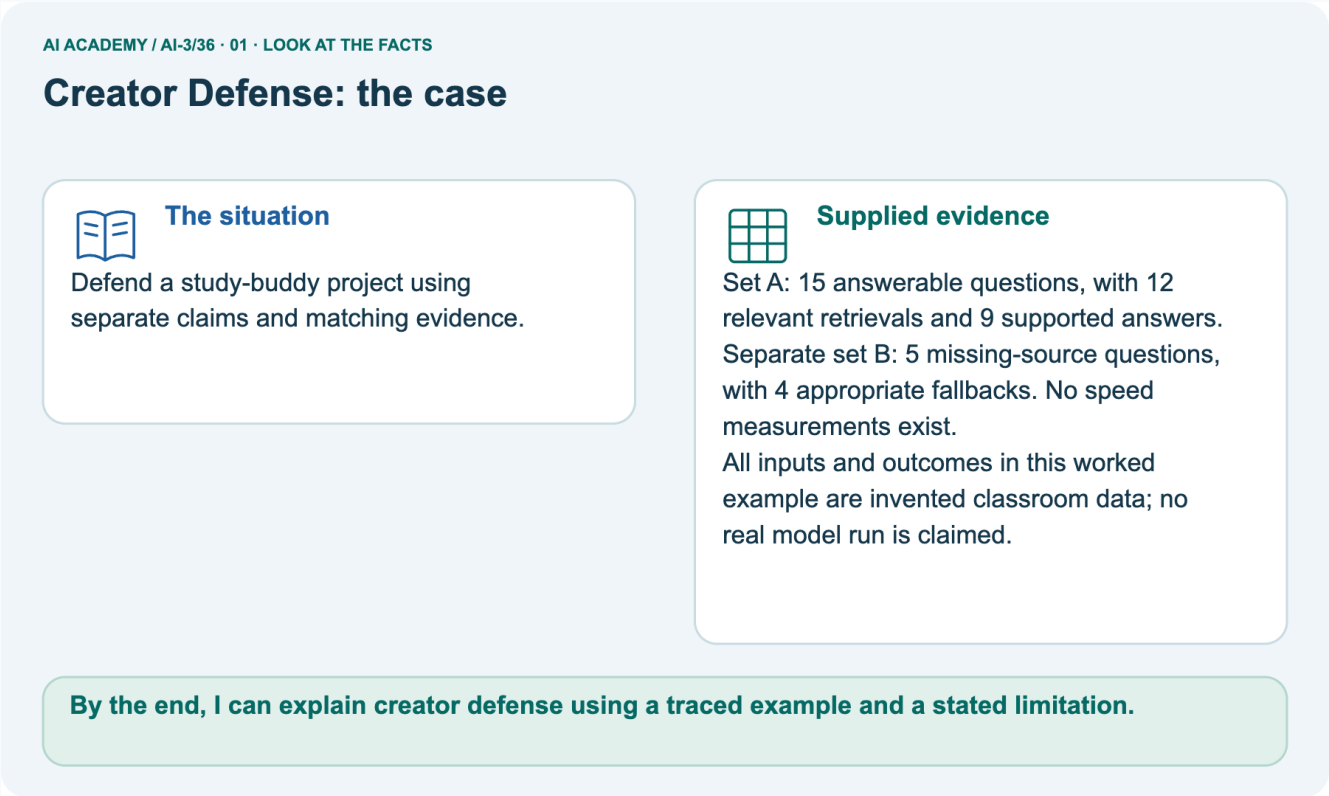
▼ Reference: the six-picture journey

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Creator Defense: the case



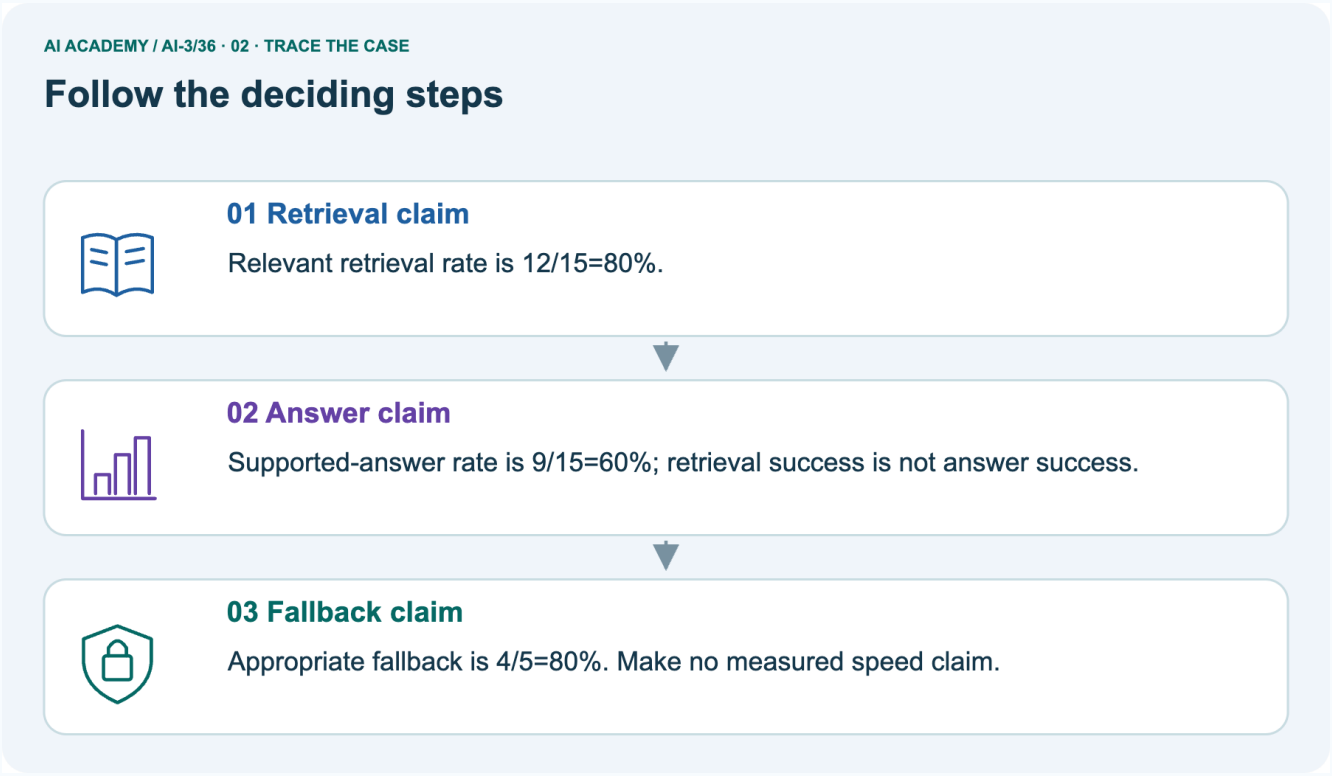
Follow the labelled picture with the matching explanation. Open the vector diagram

Defend a study-buddy project using separate claims and matching **evidence**. Supplied facts: Set A: 15 answerable questions, with 12 relevant retrievals and 9 supported answers.

Separate set B: 5 missing-source questions, with 4 appropriate fallbacks. No speed measurements exist. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



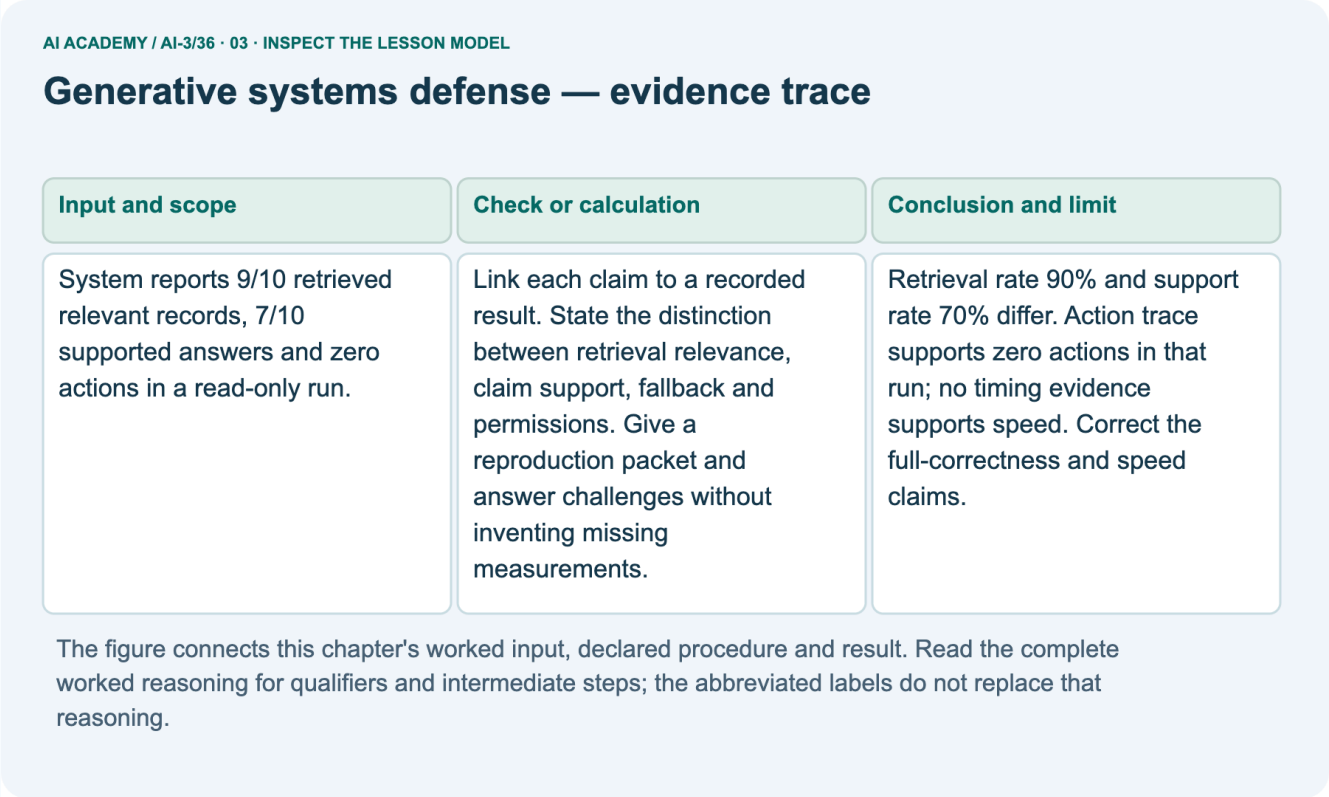
Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieval claim: Relevant retrieval rate is 12/15=**80%**. → Answer claim: Supported-answer rate is 9/15=**60%**; retrieval success is not answer success. → Fallback claim: Appropriate fallback is 4/5=**80%**.

Make no measured speed claim.

Point to each numbered stage and explain which recorded **input** it uses.

03. Generative systems defense — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Generative systems **defense** — **evidence** trace. System reports 9/10 retrieved relevant records, 7/10 supported answers and zero actions in a read-only run. | Link each claim to a recorded result.

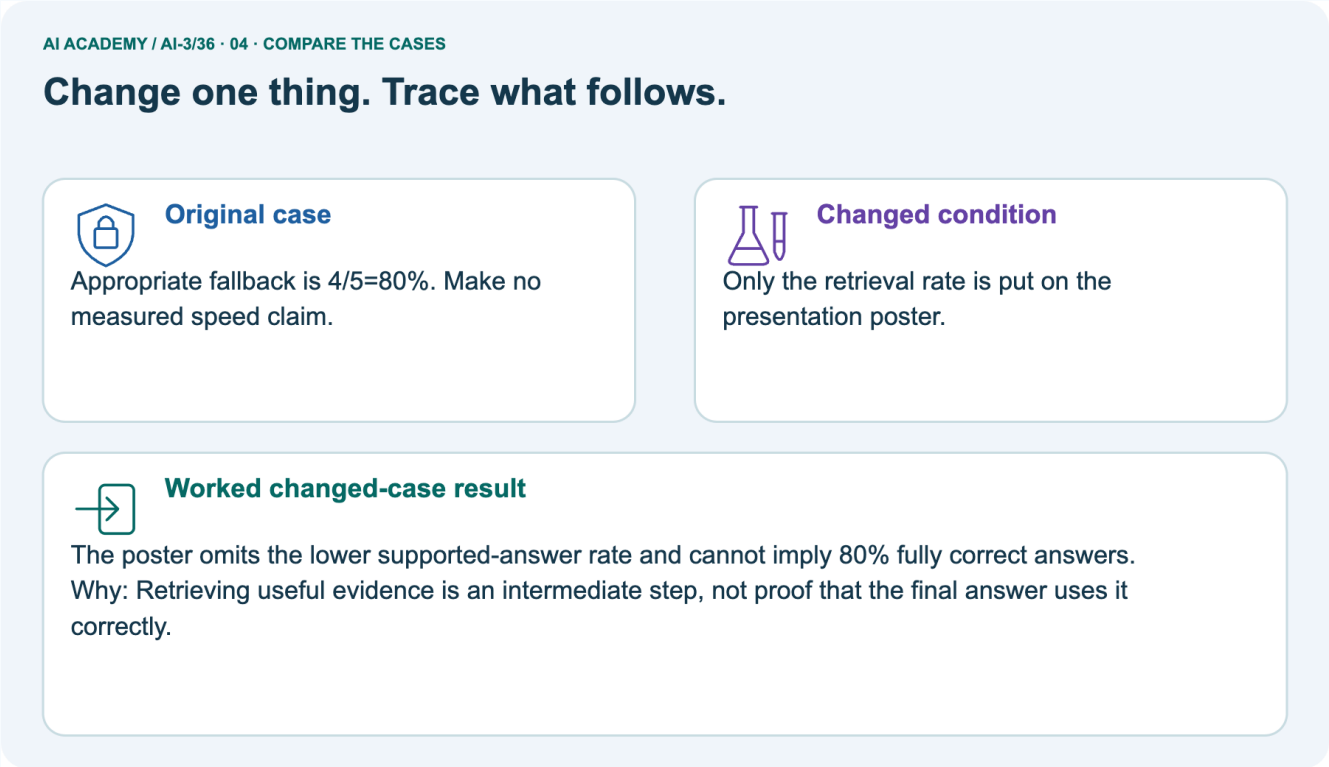
State the distinction between retrieval relevance, claim support, fallback and permissions.

Give a **reproduction packet** and answer challenges without inventing missing measurements. | Retrieval rate 90% and support rate **70%** differ. Action trace supports zero actions in that run; no timing **evidence** supports speed.

Correct the full-correctness and speed claims.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



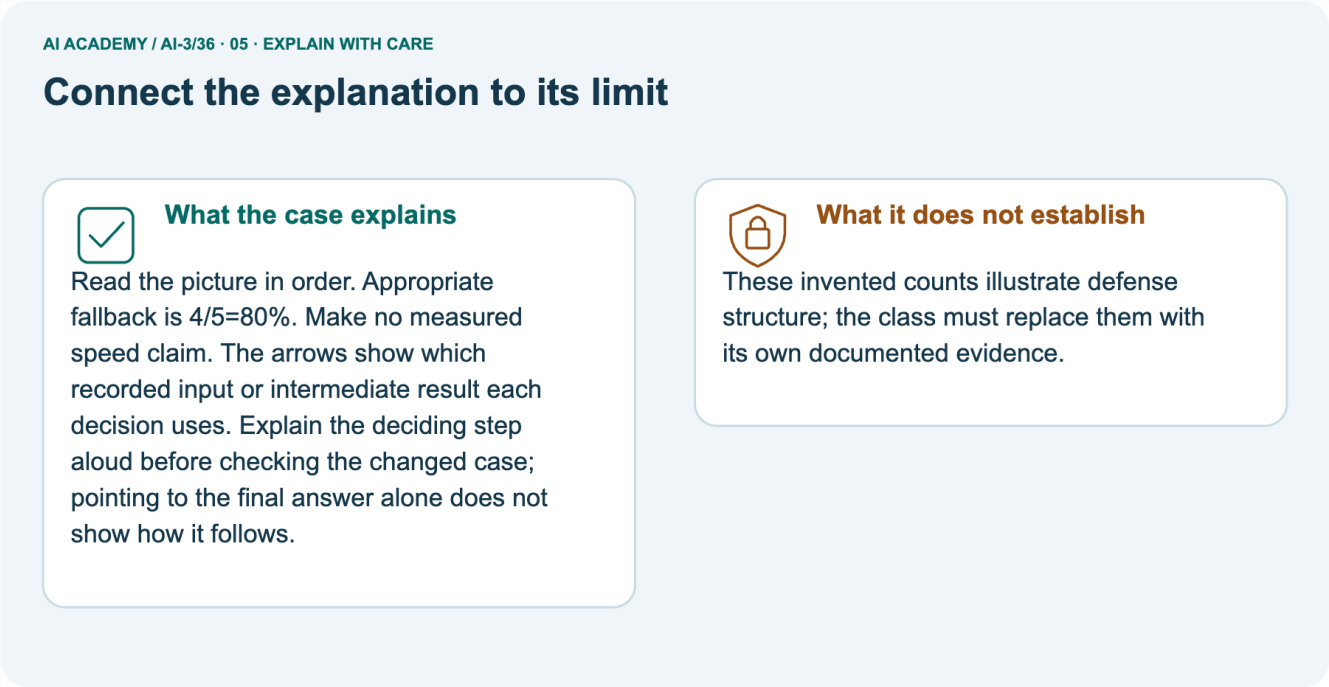
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Appropriate fallback is 4/5=**80%**. Make no measured speed claim. Changed condition: Only the retrieval rate is put on the presentation poster.

Expected result: The poster omits the lower supported-answer rate and cannot imply **80%** fully correct answers. Reason: Retrieving useful **evidence** is an intermediate step, not proof that the final answer uses it correctly.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

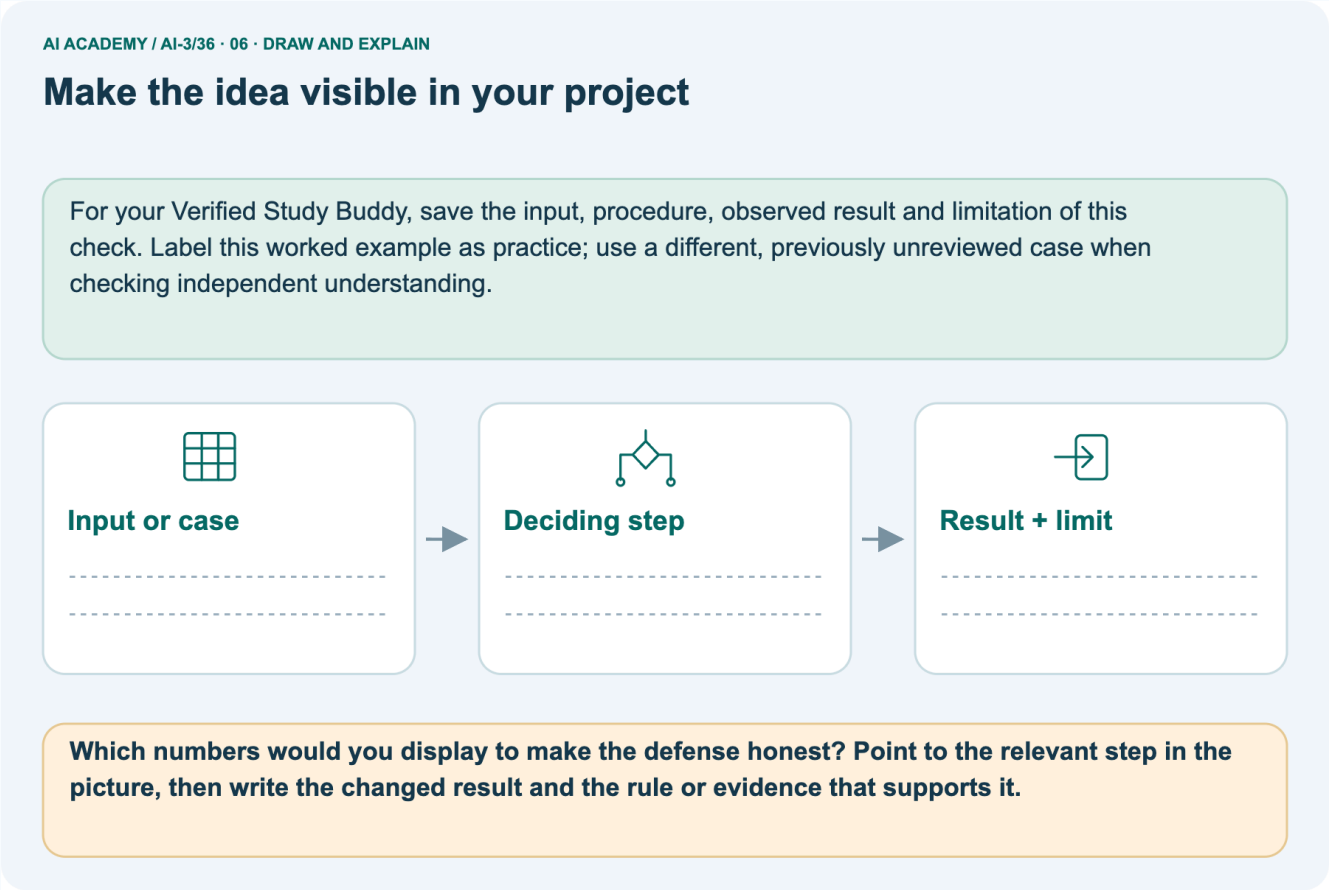
Read the picture in order. Appropriate fallback is 4/5=**80%**. Make no measured speed claim. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: These invented counts illustrate **defense** structure; the class must replace them with its own documented **evidence**.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which numbers would you display to make the **defense** honest? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Creator Defense: the case

02. Follow the deciding steps

03. Generative systems defense — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.